



Dataset «HousePrice»



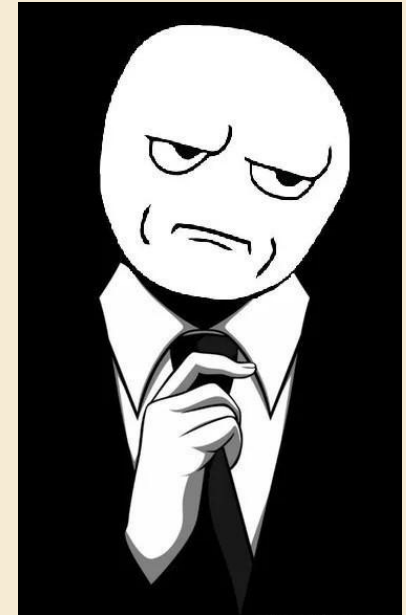
TEAM PROJECT



- Sen Gvan



- Denis

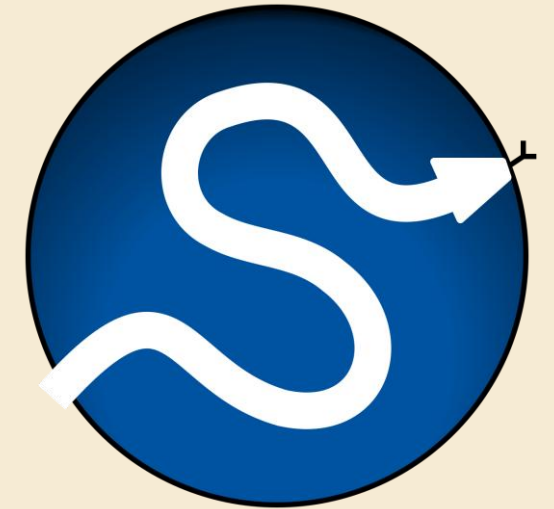


- Andrey

LIBRARY



Matplotlib



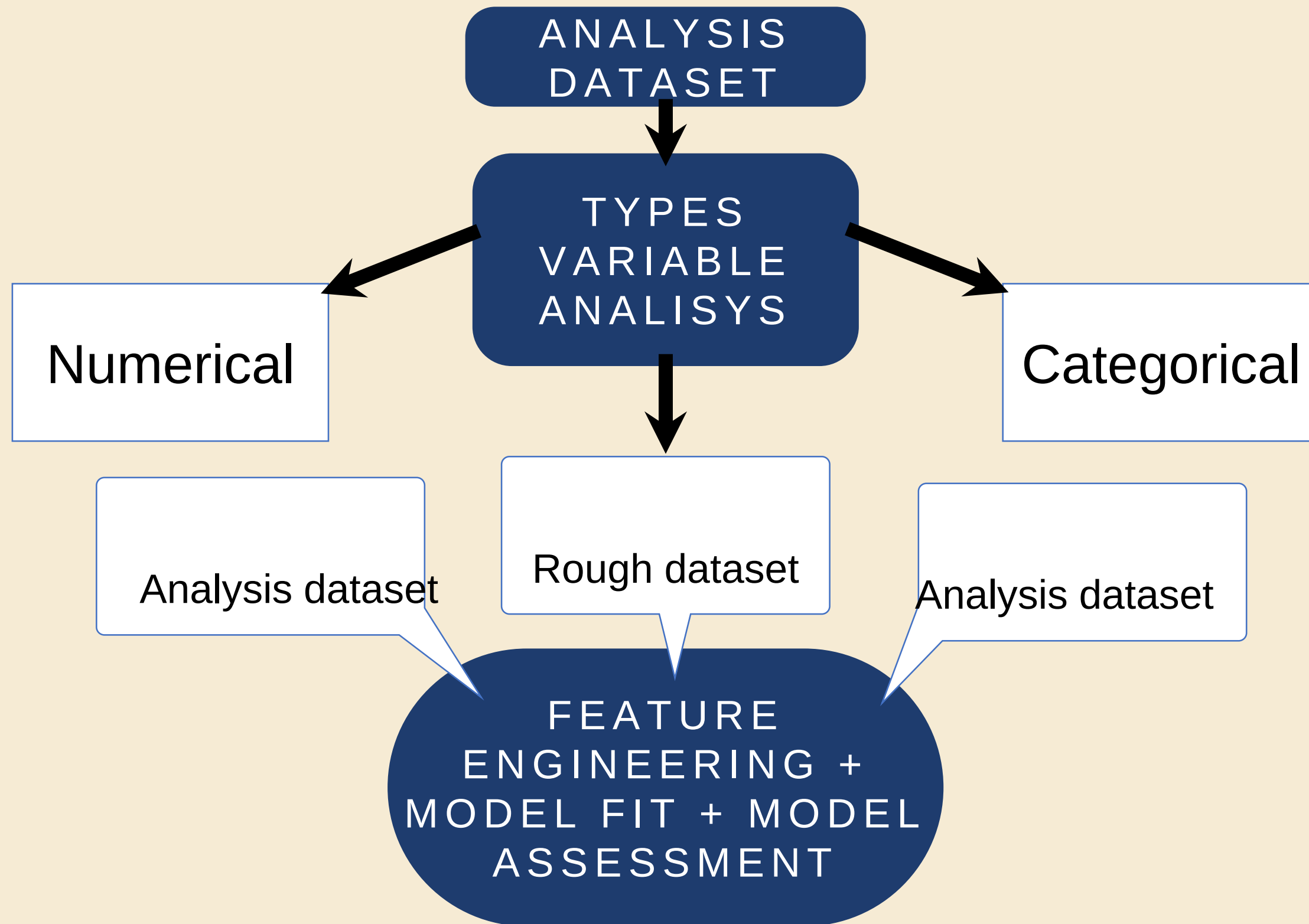
SCIPY



seaborn



FEATURE
ENGINE



PROJECT OBJECTIVE

1. Understand and define the problem.
2. Define the requirements for solving the problem.
3. Data collection and preparation.
4. Analysis. Exploratory Data Analysis (EDA) of variables.
5. Feature engineering
6. Building multiple models.
7. Model evaluation and selection of the best one



Hypotheses

1. The price of a house depends on the overall quality (Overall quality)
2. The price of the house depends on the living area (GrLivArea)
3. The price of the house depends on the year of construction (YearBuilt)
4. The price of a house depends on the type of housing (MSSubClass)
5. The price of a house depends on the (Neighborhood)



General dataset Information

The dataset contains information on 1,460 houses sold in the city of Ames, Iowa, USA

Housing sales data in Ames from 2006 to 2010.

1,460 properties with 80 features for each real estate unit.



Key Factors Affecting Price

1

GrLivArea

An increase in price by \$100 per square foot.

2

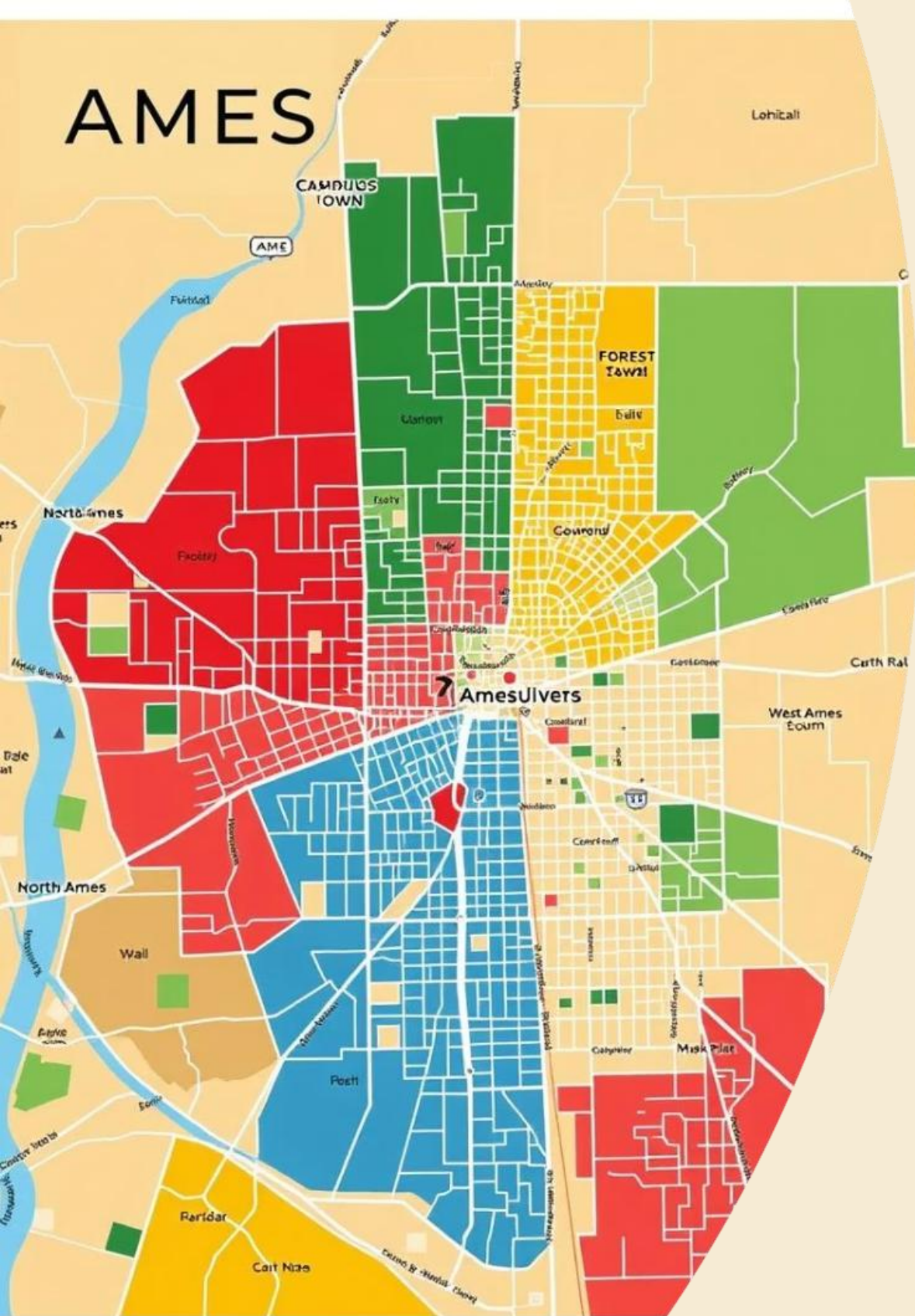
OverallQual

Each point adds \$20,000 to the value.

3

Neighborhood

StoneBr is approximately 30% more expensive than NAmes.



Ames District: Example



North Ames (NAMES)

An old, well-established neighborhood with developed infrastructure



Iowa State University (IDOTRR)

Located near the university, popular among students



Stone Brook (StoneBr)

Modern homes with high prices and quality construction.



Feature Engineering

1

Split on 3 Datasets

- Rough dataset
- Top 15 dataset
- Analysis dataset

2

Impute missing values

- on 0
- KNN

3

Encoding

- OneHotEncoder
- Ordinal Encoder

4

Transformation and discretisation

- Yeo-johnson
- Logarithmic
- Manual discretisation

5

Scaling

- Robust
- Standard

Algorithms ML

1.Linear regression

2.SVR

3.Lasso

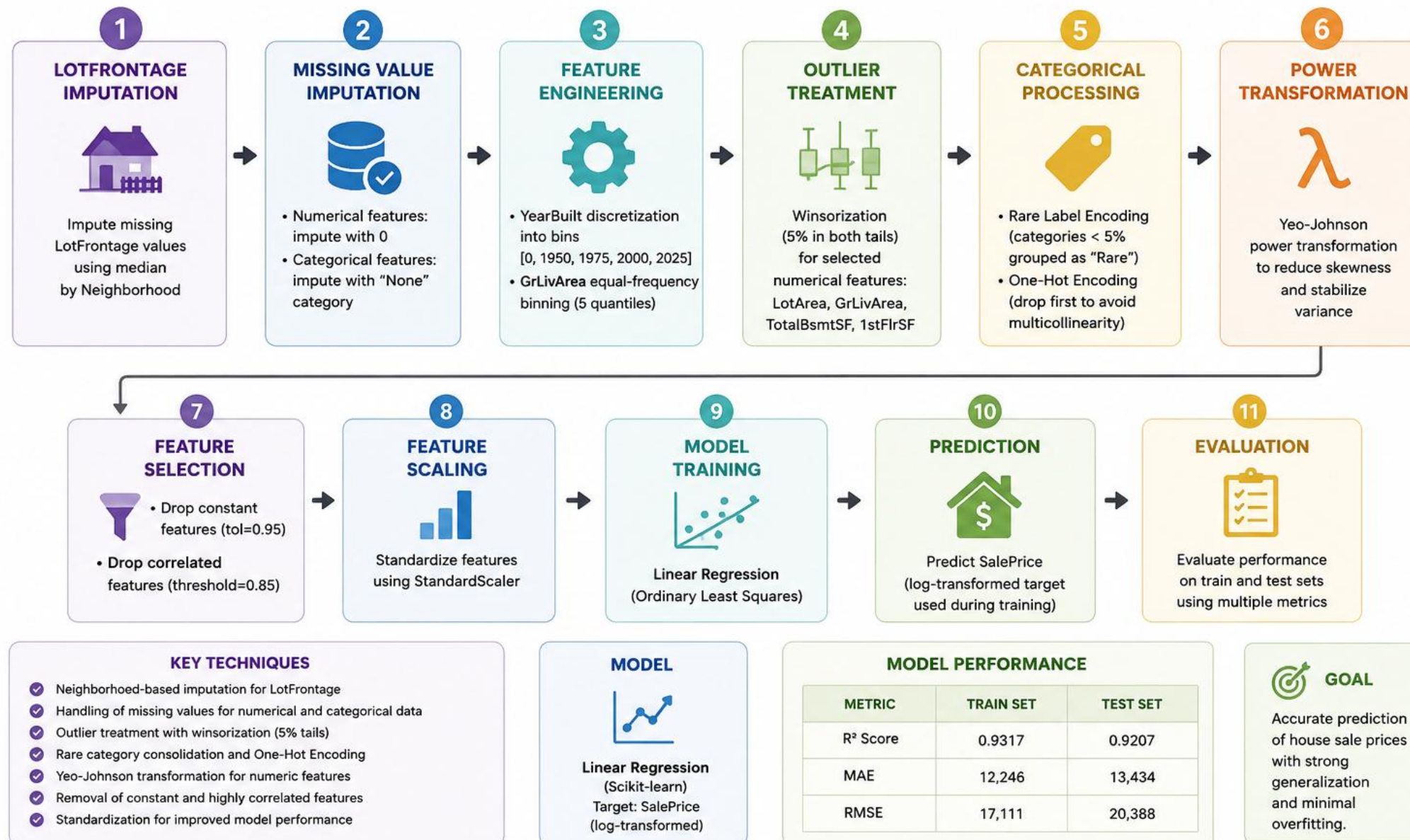
4.Random Forest

5.GradientBoosting (XGBoost, GBM)

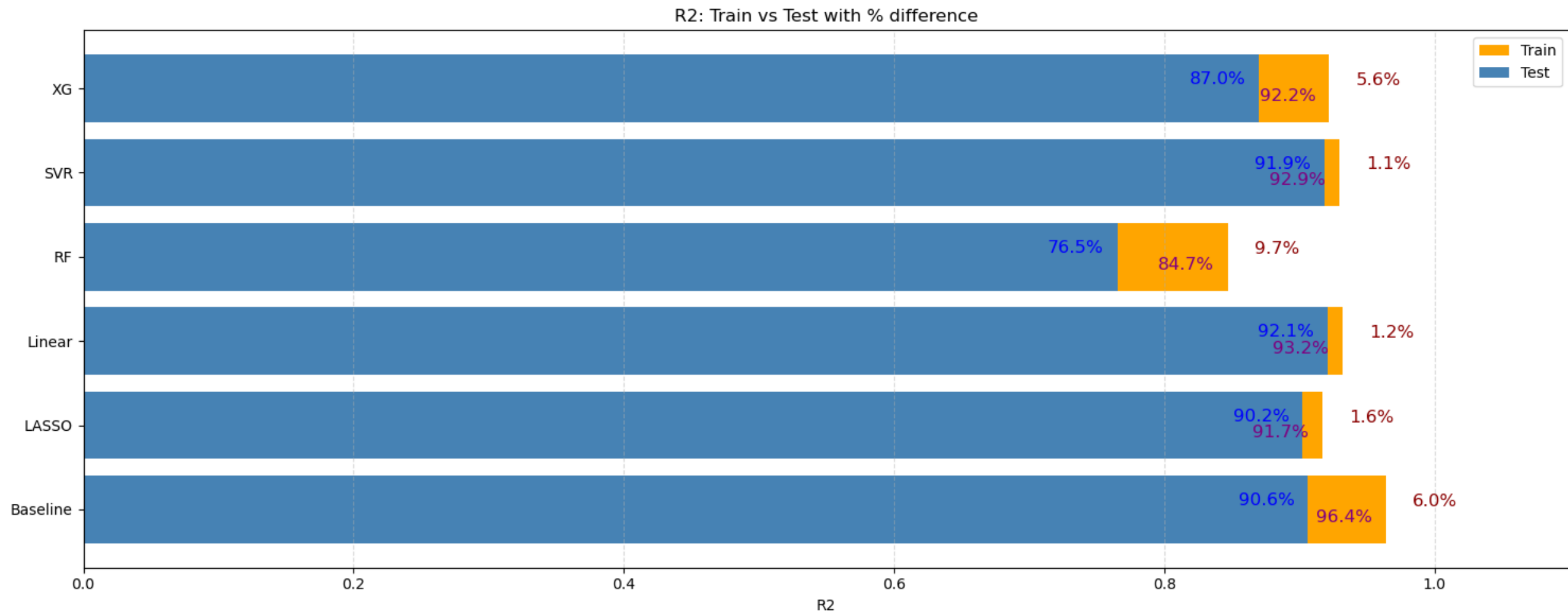
Results

HOUSE PRICE PREDICTION PIPELINE

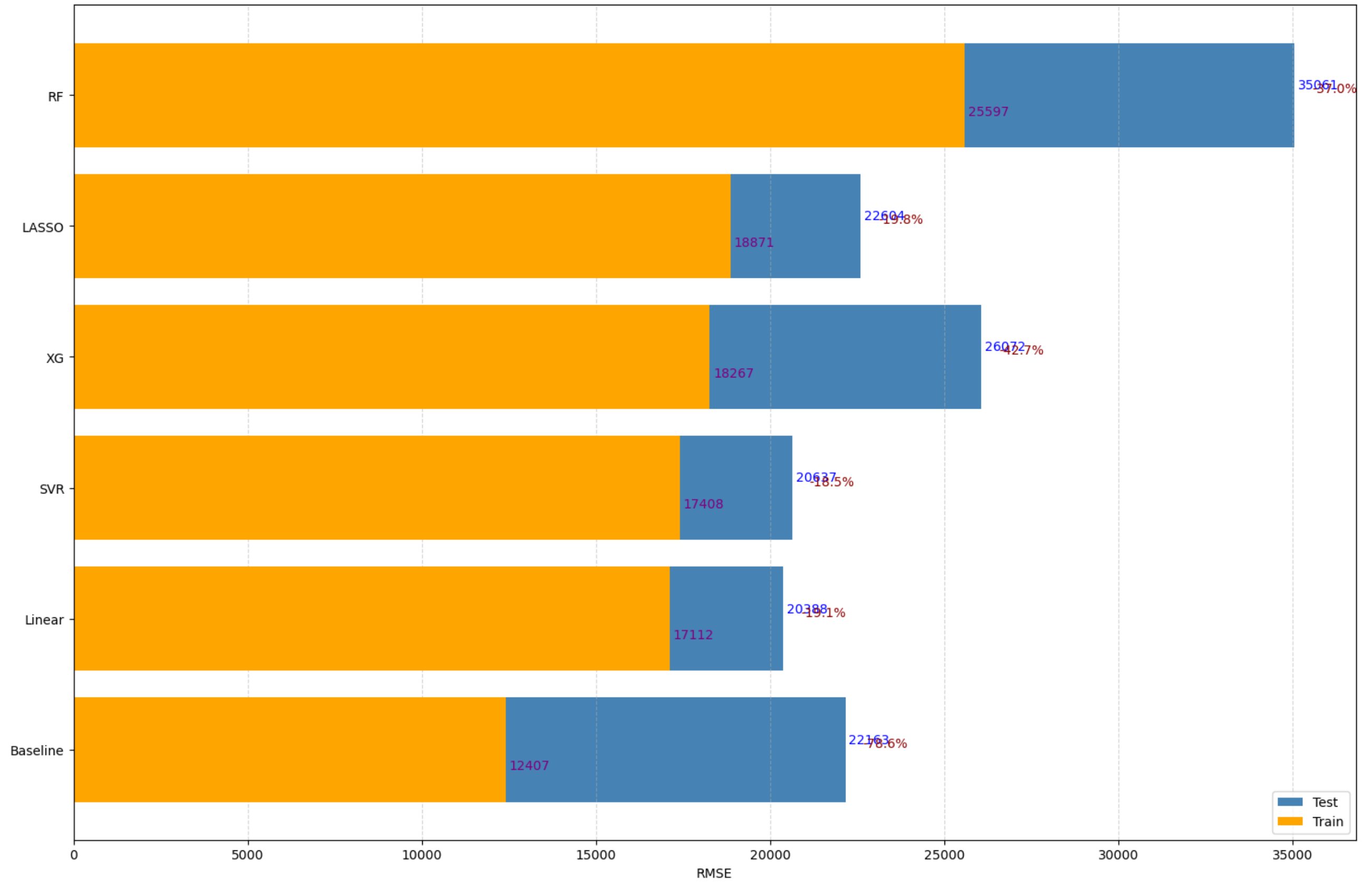
End-to-end data preprocessing and modeling workflow



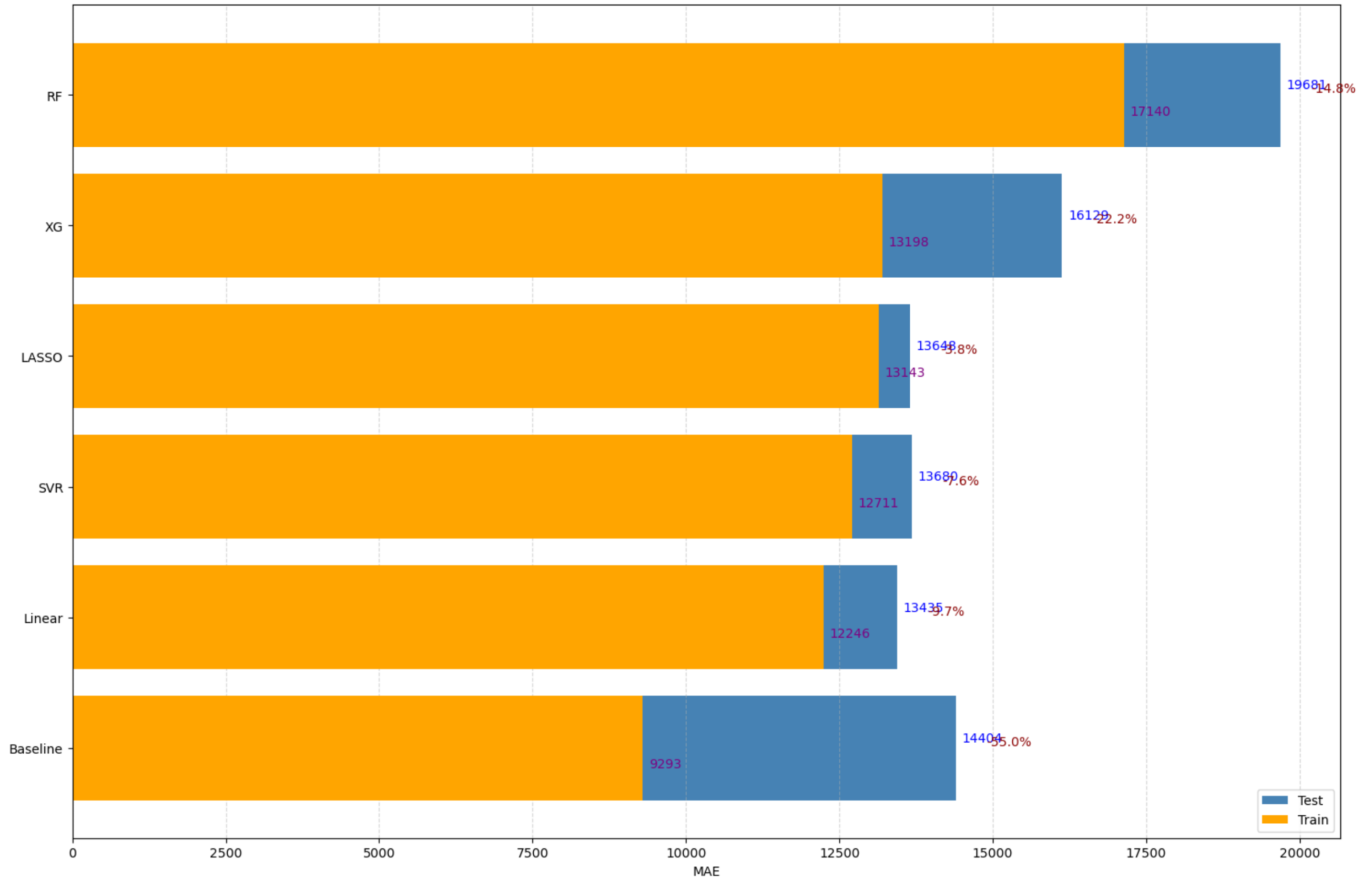
RoughDataset



RMSE: Train vs Test with % difference

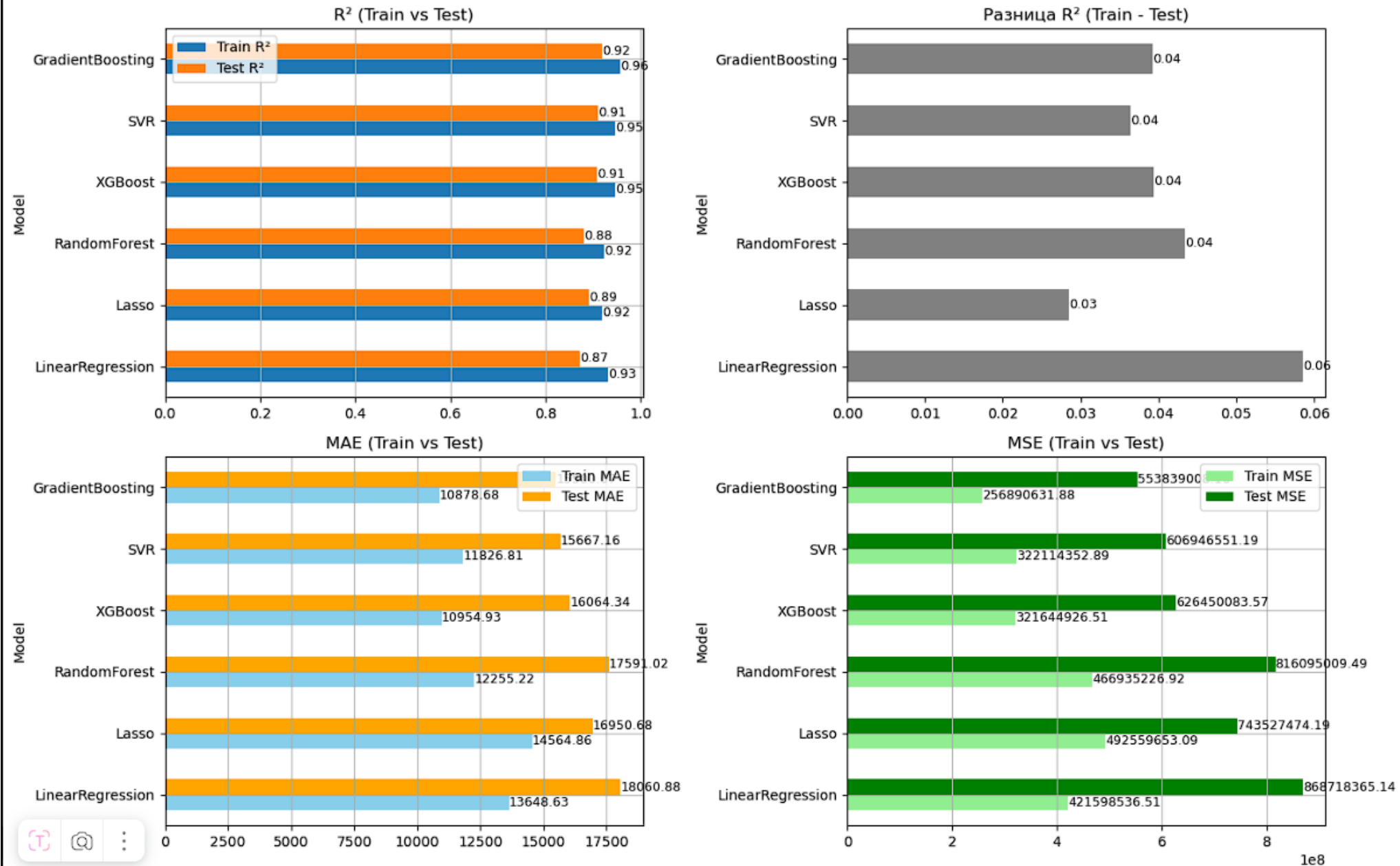


MAE: Train vs Test with % difference



Top15 importance

Metrics to compare

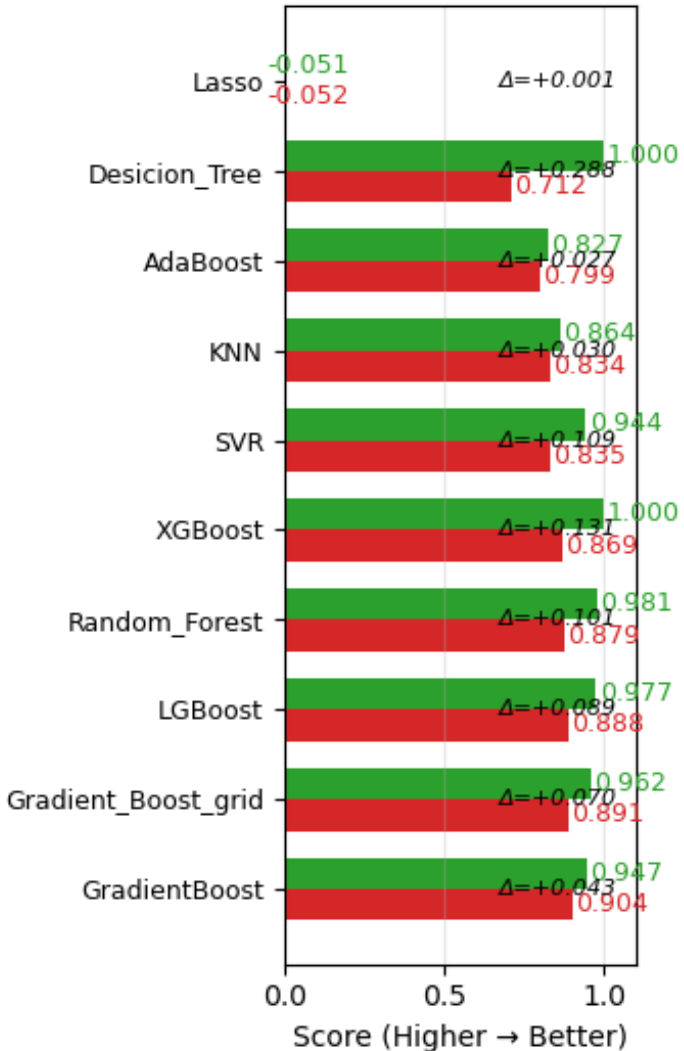


Analysis dataset

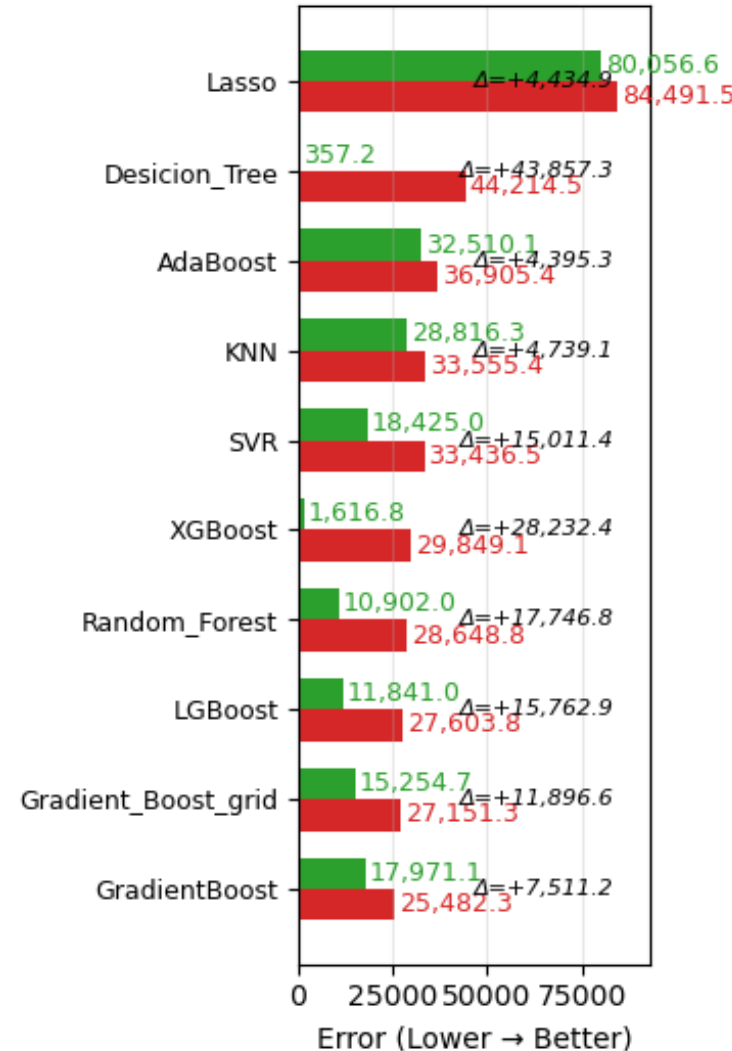
Model Performance Comparison

Train Test

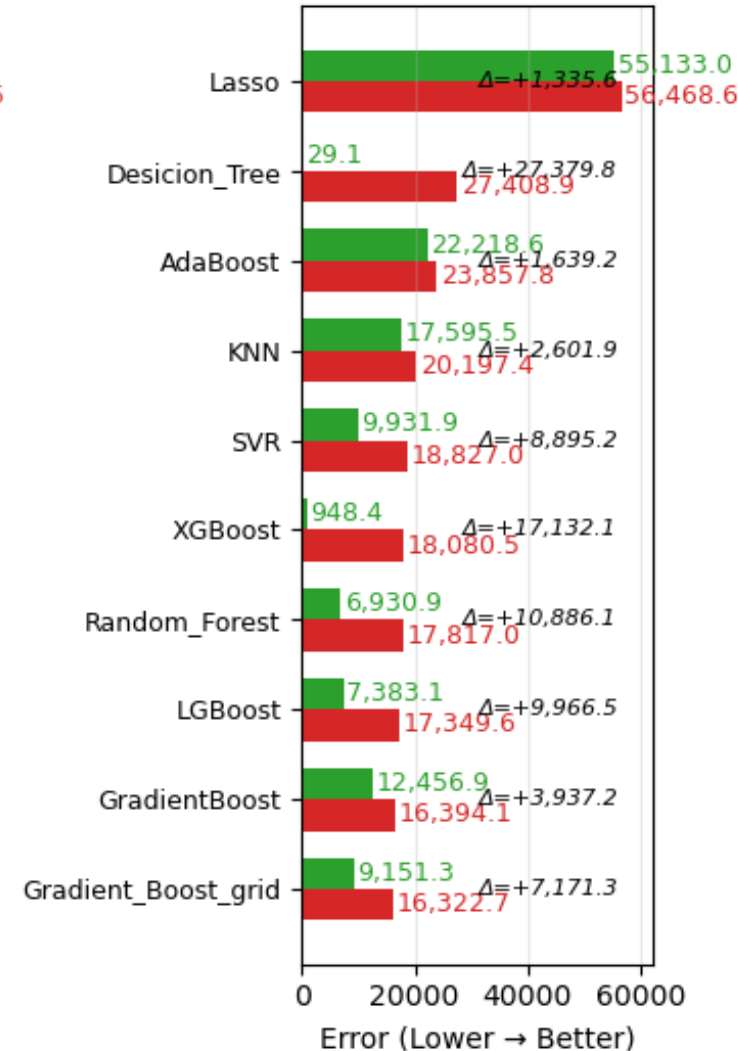
R2



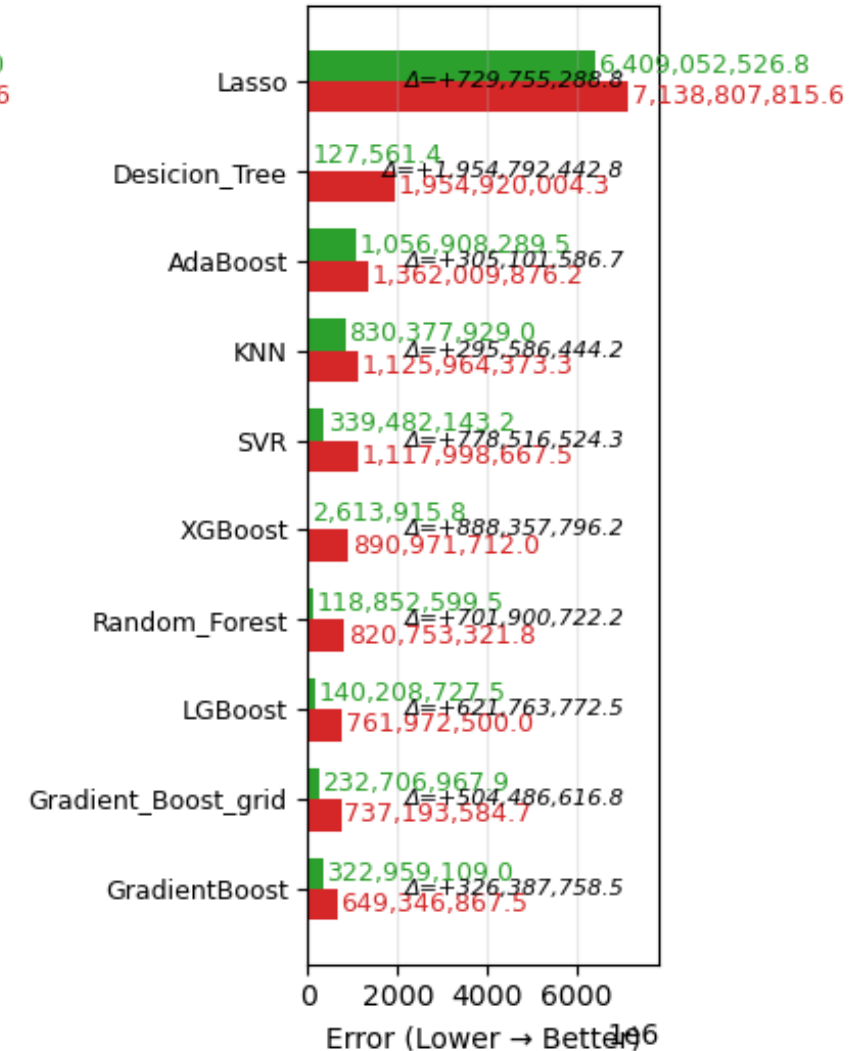
RMSE



MAE



MSE

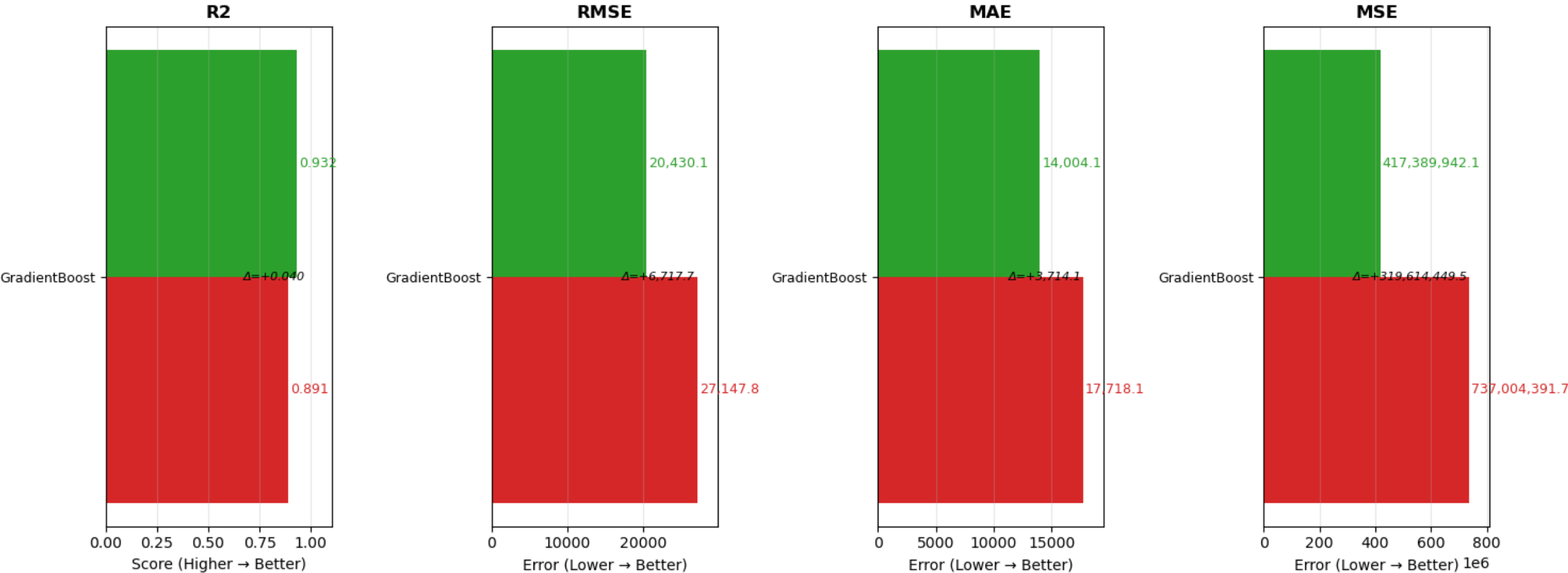


Correlation dataset

40

Model Performance Comparison

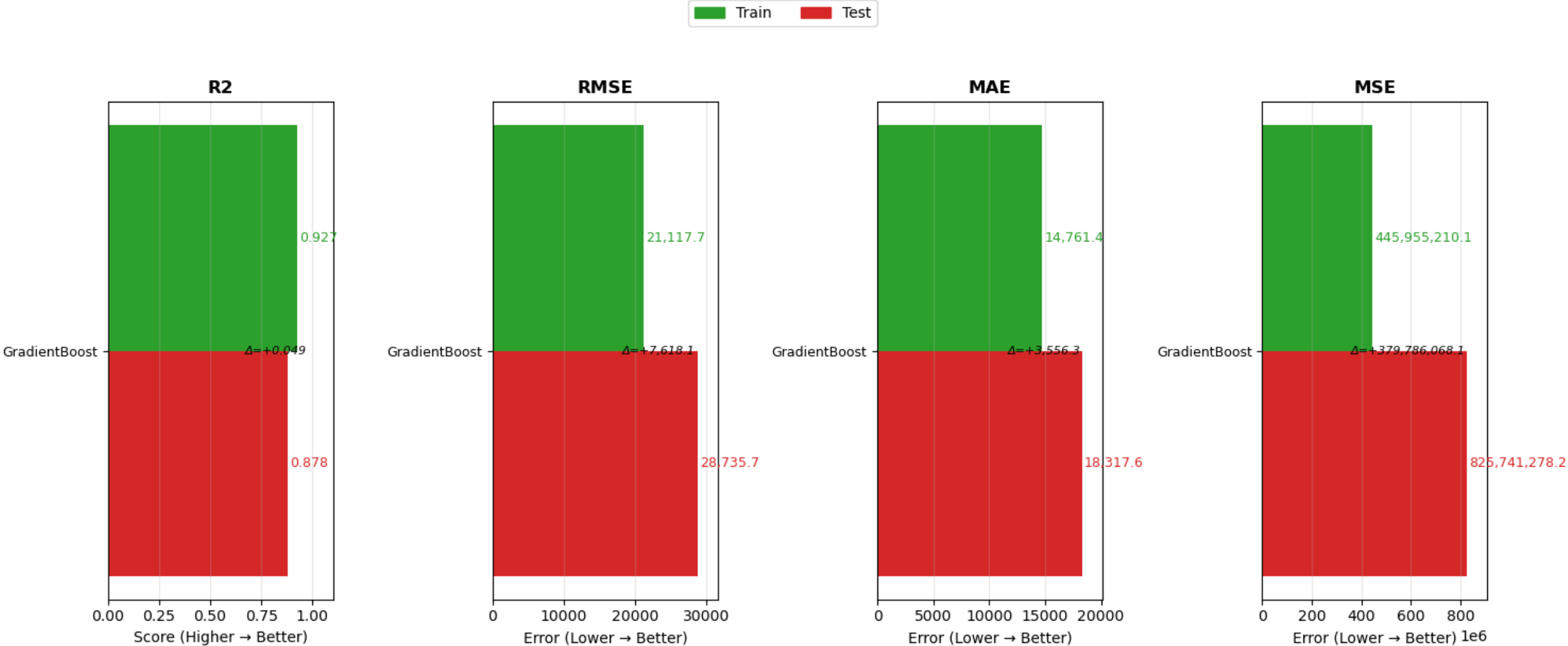
Train Test



Correlation dataset

50

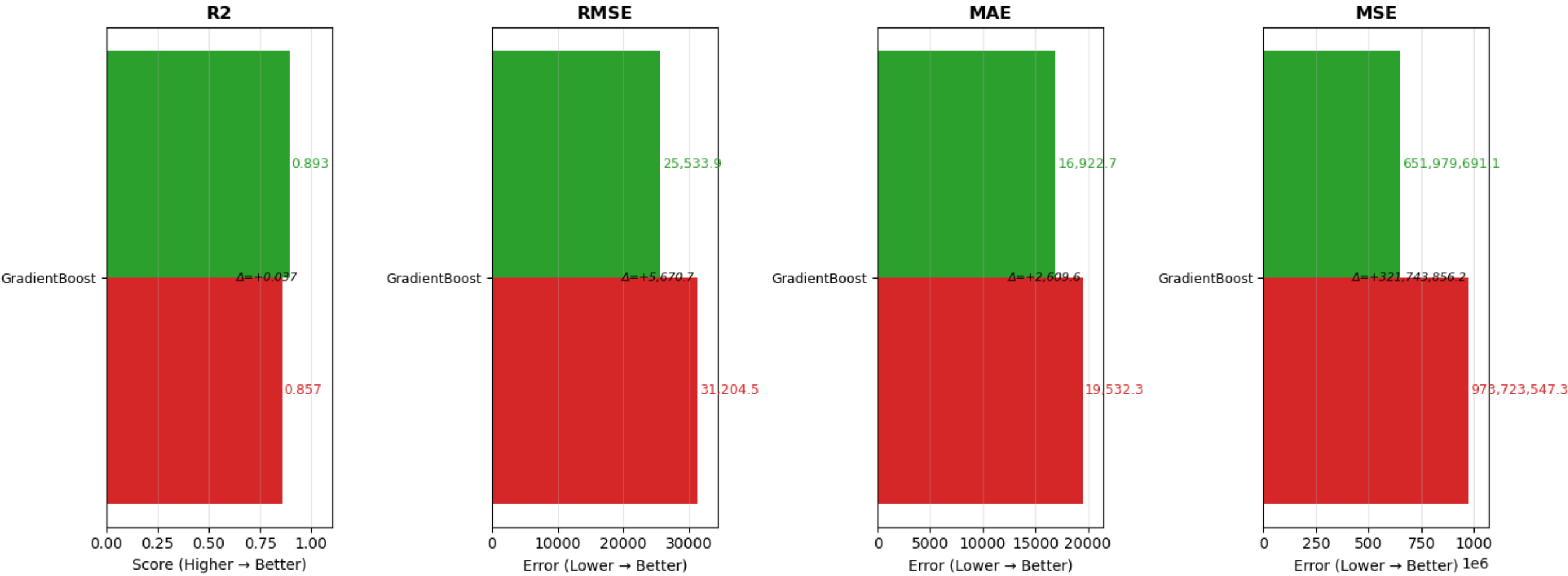
Model Performance Comparison



Correlation dataset

60

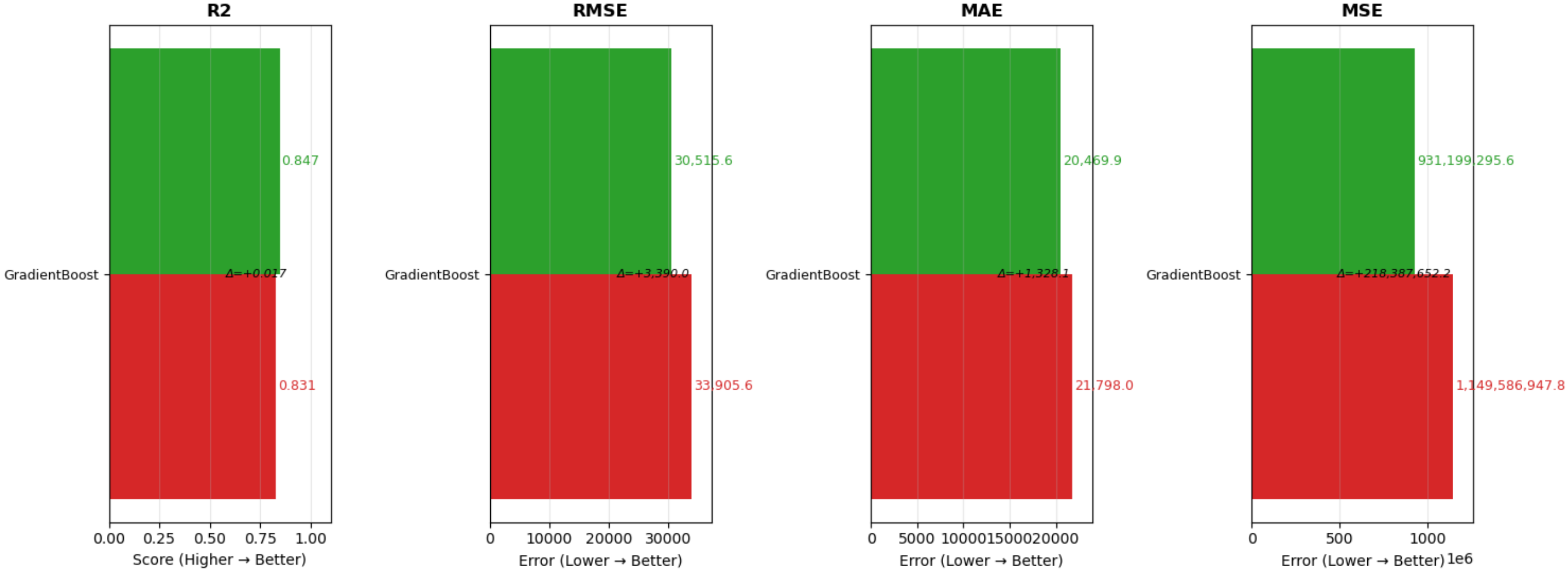
Model Performance Comparison



Correlation dataset

65

Model Performance Comparison





Conclusion

An important factor is analysis!

The **Linear Regression** model achieved the best performance among the tested approaches. Its metrics demonstrate both high accuracy and strong generalization ability.

Training Set:

- $R^2 = 0.9317$
- $MAE = 12,246$
- $RMSE = 17,111$

Test Set:

- $R^2 = 0.9207$
- $MAE = 13,434$
- $RMSE = 20,388$